FACE-VERIFIED EYE AND HAND CONTROLLED COMPUTER INTERFACE

Under the guidance of Dr. SHAIK RIYAZ HUSSAIN SIR
Team members:
P.Naveen Kumar -N200410
K.Santhosh Kumar -N200922
B.Usha Rani -N200315

RAJIV GANDHI UNIVERSITY OF KNOWLEDGE TECHNOGIES, NUZVID

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ABSTRACT

Abstract

- An Eye-Controlled Mouse Cursor is an assistive technology solution that allows individuals.
- Especially those with physical disabilities affecting limb movement.
- To control the movement of a computer cursor using just their eye movements.
- Also system volume control using hand gestures and face recognition for authentication

INTRODUCTION

Introduction

- An algorithm for controlling the movement of a computer screen cursor using the iris movement of the eye.
- As human-computer interactions are evolving. Physically disabled people face challenges using traditional input devices.
- The proposed project provides an affordable and practical solution.
- This project explores the control of a cursor using eye movements.
- And also this project explores the secure face recognition of the person for courser control and volume control using hand gestures.

PROBLEM STATEMENT

problem statement

- To create a hands-free system that enables physically disabled individuals to control a mouse pointer using eye movements and blinks, improving their accessibility and interaction with computers.
- Traditional HCI methods (e.g. speech recognition, EEG, EMG) are expensive or impractical.
- Physically disabled people need a hands-free and cost-effective solution.
- As current infrared-based eye tracking systems are costly.

PROBLEM SOLUTION

Problem solution

- A system using only a webcam and PYTHON Open CV based software.
- First it recognizes the person face and then grant access to the computer to control the cursor [1].
- Tracks iris movement to control the mouse cursor. Includes a blinking mechanism for clicking.
- No need for expensive sensors or external hardware. The proposed system is a virtual interactive module for the users and can be used as an alternative module of touch screen technology.

Block Diagram

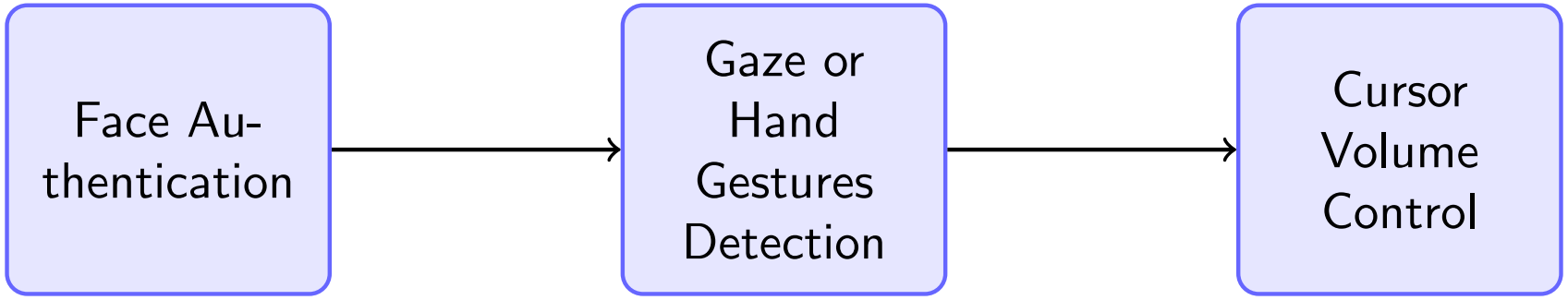

Figure: Face Authentication and Control System.

Vision based cursor control

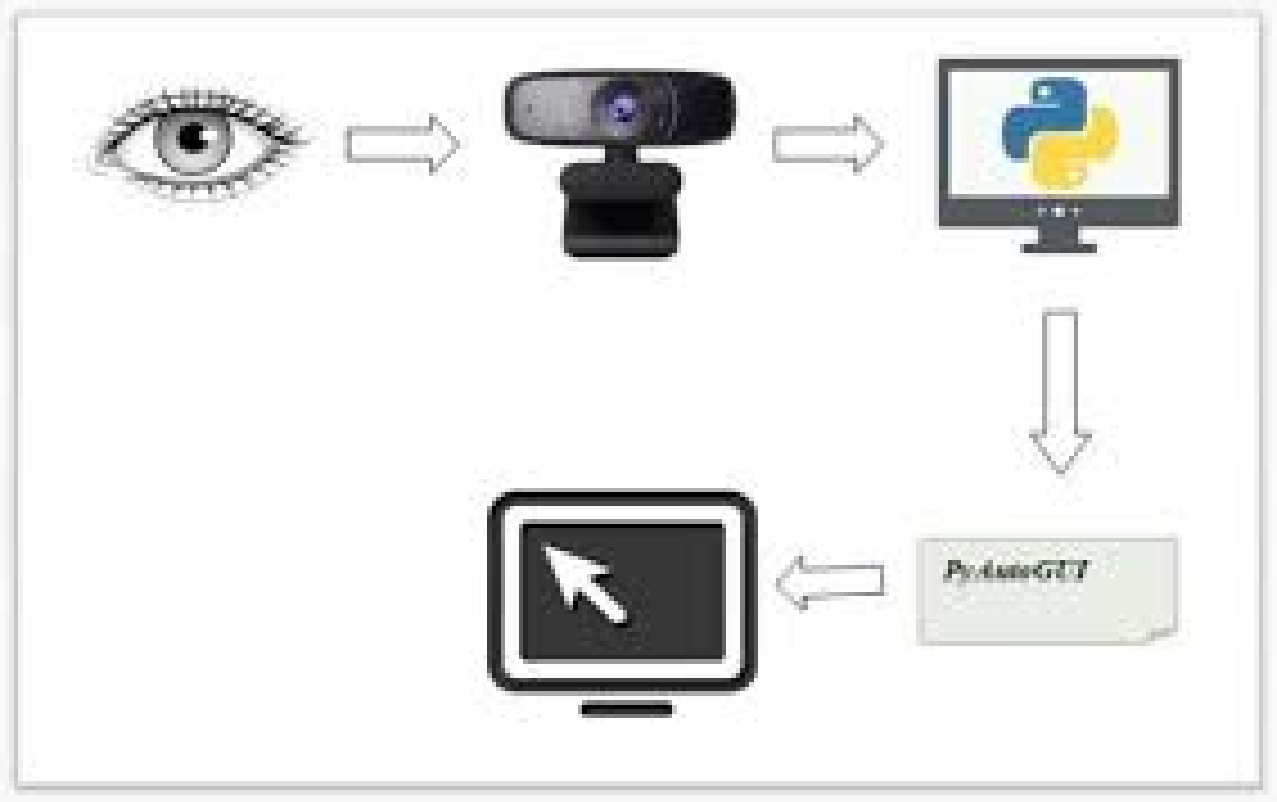

Figure: Gaze detection and then cursor control

OBJECTIVE

Objective

- To develop a system for cursor control using an eye-tracking mechanism aimed at helping physically challenged individuals.
- In this project, a computer application is developed using Python and OpenCV.
- This application will make it possible for the user to move the cursor point on the computer screen and also additional functions like clicking on the computer screen.

SYSTEM COMPONENTS

software

- Eye Tracking Algorithm: Detects and tracks eye position and movement[1].
- Face recognition algorithm :Detects and scans the face and allows only the recognized faces which were added by the owner[4].
- Cursor Control Module: Maps eye movement coordinates to cursor position.
- Click Detection: Uses blinks or dwell time (hovering over a spot) to perform clicks[3].

Hardware

- Camera (Webcam or Infrared Camera): Captures real-time images or video of the user's eyes.
- Computer/Processing Unit: Processes the video input and translates eye movements into cursor actions.
- (Optional) Infrared Light Source: For better eye tracking in low light or precise tracking.

TECHNOLOGIES USED

Languages and libraries

- Languages: Python (popular for computer vision and prototyping)
- Libraries:
- OpenCV – for image/video processing.
- Deep Face - for face recognition.
- Dlib or MediaPipe – for facial and eye landmark detection
- PyAutoGUI – for cursor control
- pycaw – for volume control
- Machine Learning (Optional):
- For more precise eye movement prediction and calibration

BASIC WORKING PRINCIPLE

Basic Working Principle

- **Face Detection**: Capture the user's face and allow access to control cursor for saved recognized faces only.
- **Eye Detection**: Capture the user's face and eyes using a webcam.
- **Pupil Tracking**: Track the pupil movement across the screen.
- **Mapping**: Translate eye position into screen coordinates.
- **Click Events**:
  - Use **blink detection** (Right Wink = Right Click).
  - (Left Wink = Left Click).
- **Hand gesture:** Volume control using Hand Gestures.[2]

OUTPUT

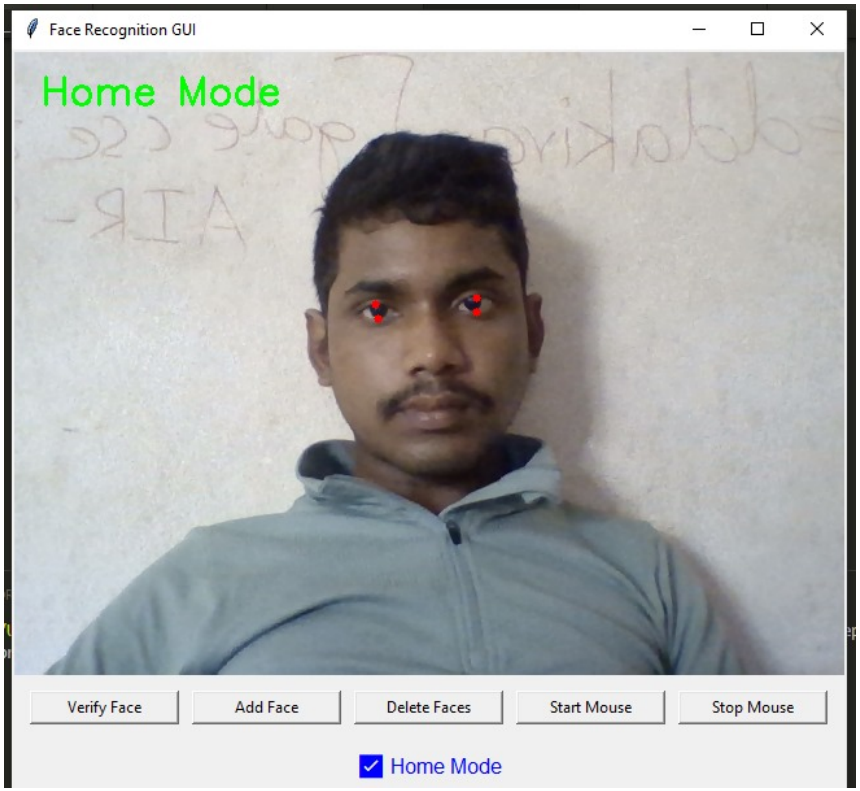

Figure: Home Mode

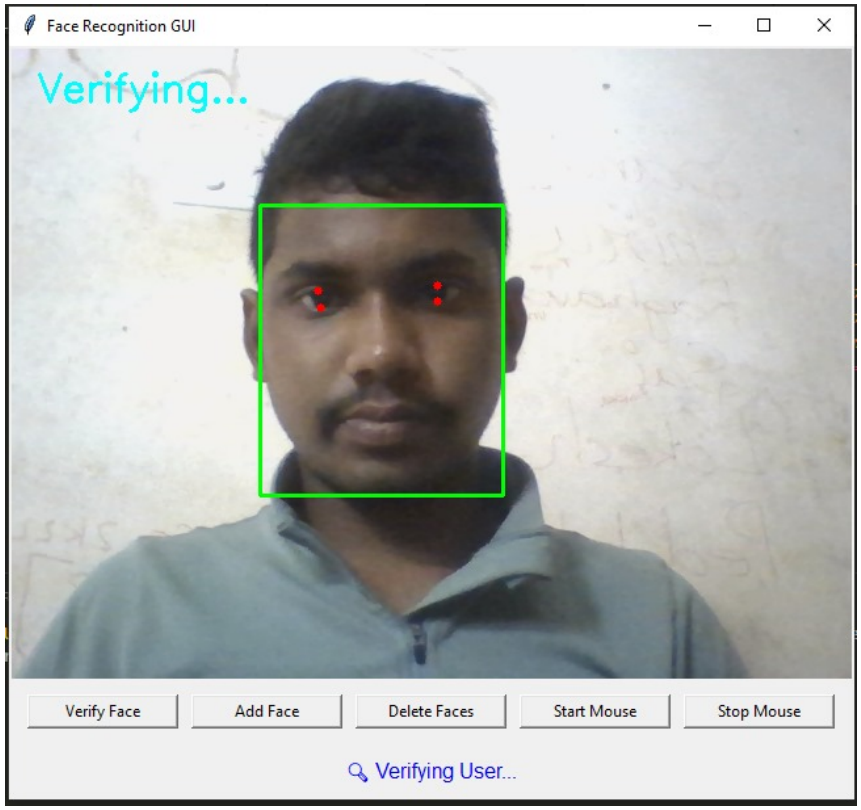

Figure: Verification Mode

OUTPUT

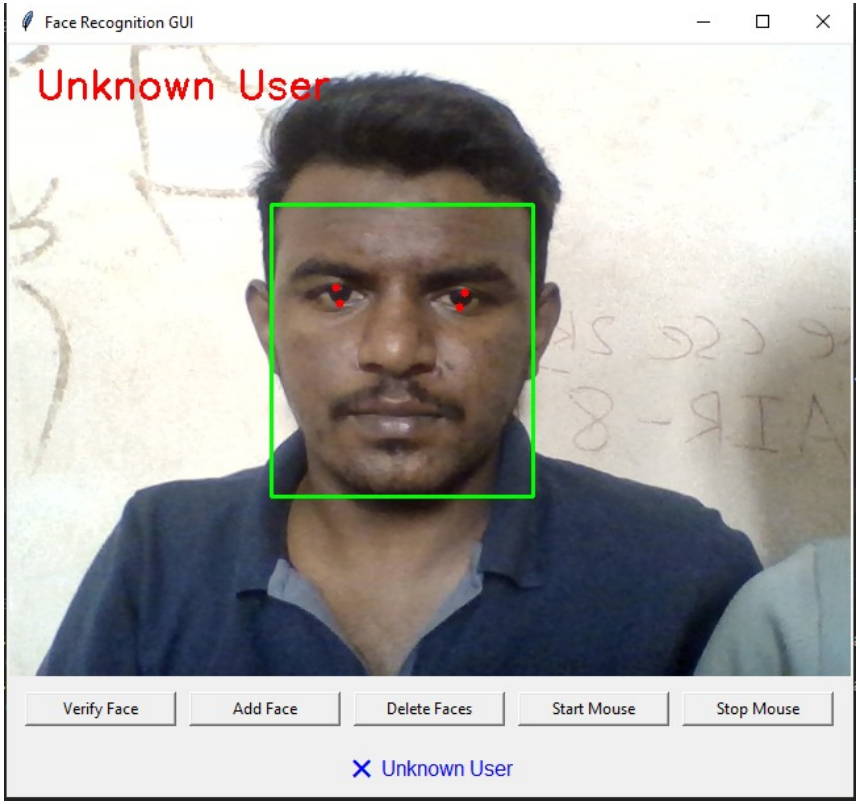

Figure: Home Mode

Figure

OUTPUT

Figure: Gesture Volume Controlling

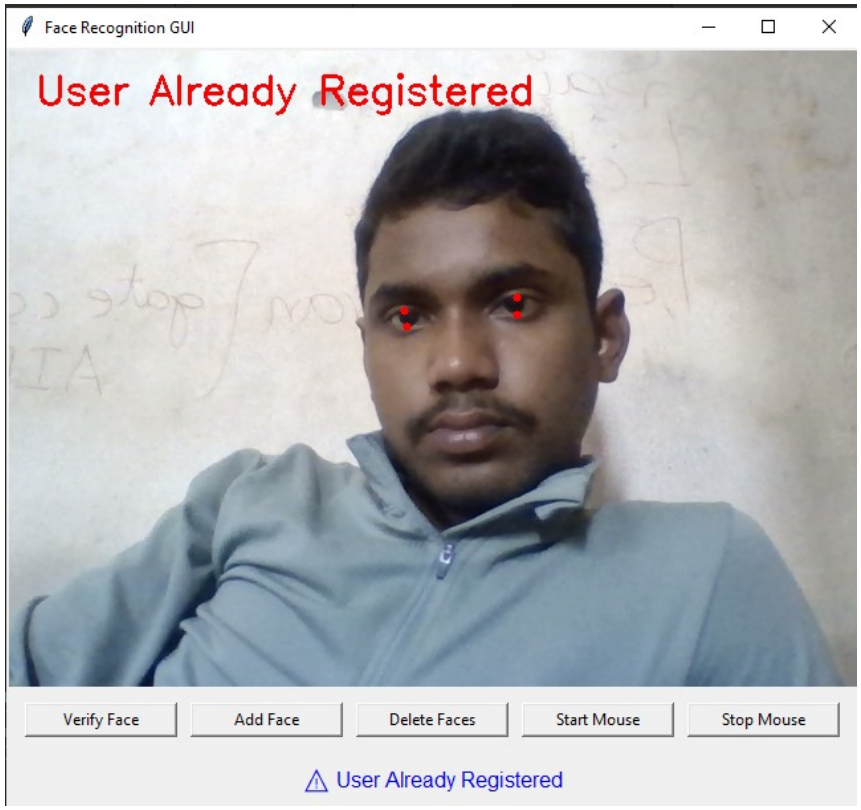

Figure: New User Registration

APPLICATIONS

Applications

- Perform actions like move the cursor in all directions, scroll, drag and click without physical interaction.
- Computer interaction for physically disabled individuals.
- Home automation (controlling lights, TV, doors).
- Gaming and virtual reality applications.

CONCLUSION AND FUTURE WORK

Conclusion

- A low-cost, webcam-based eye-controlled cursor was developed.
- It can be successfully implemented and tested by using python Open CV
- Future Improvements:
- Enhancing up/down movement accuracy.
- Increasing processing speed for real-time interaction. Integrating with other assistive technologies.

REFERENCE

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Thank you!